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ELECTRONIC CIGARETTE

Abstract

An electronic cigarette, including: a main body, the main body including a storage chamber for storing atomizable substance and an atomization core for heating the atomizable substance. The main body further includes a first end including a first channel for guiding and injection of the atomizable substance, and a second end including a second channel for guiding and injection of the atomizable substance. The first end is opposite to the second end. The first channel includes a first injection port for the atomizable substance and the second channel includes a second injection port for the atomizable substance. Both the first injection port and the second injection port communicate with the storage chamber; a first sealing film is disposed in the first injection port and a second sealing film is disposed in the second injection port. The first sealing film and the second sealing film are puncturable.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Pursuant to 35 U.S.C. § 119 and the Paris Convention Treaty, this application claims foreign priority to Chinese Patent Application No. 202420470525.0 filed Mar. 11, 2024, the contents of which, including any intervening amendments thereto, are incorporated herein by reference. Inquiries from the public to applicants or assignees concerning this document or the related applications should be directed to: Matthias Scholl P. C., Attn.: Dr. Matthias Scholl Esq., 245 First Street, 18th Floor, Cambridge, MA 02142.

BACKGROUND

[0002] The disclosure relates to an electronic cigarette compatible with different types of automated filling machines.

[0003] Conventional open-type e-cigarettes or atomizers usually have an e-liquid injection hole, and a detachable sealing plug is inserted in the hole. During e-liquid filling, the sealing plug is manually opened. The filling mode cannot meet the needs of automated operation.

[0004] Existing automated filling machines on the market only have one e-liquid injection mode, either top-down injection or bottom-up injection. However, existing electronic cigarettes or atomizers usually only have one feeding inlet disposed on the upper end or the lower end, which cannot be simultaneously compatible with different types of automated filling machines.

SUMMARY

[0005] The disclosure provides an electronic cigarette that can be compatible with different types of automated filling machines.

[0006] Specifically, the disclosure provides an electronic cigarette comprising a main body, the main body comprising a storage chamber for storing atomizable substance and an atomization core for heating the atomizable substance. The main body further comprises a first end comprising a first channel for guiding and injection of the atomizable substance, and a second end comprising a second channel for guiding and injection of the atomizable substance; the first end is opposite to the second end; the first channel comprises a first injection port for the atomizable substance and the second channel comprises a second injection port for the atomizable substance; both the first injection port and the second injection port communicate with the storage chamber; a first sealing film is disposed in the first injection port and a second sealing film is disposed in the second injection port, and the first sealing film and the second sealing film are puncturable and able to restore to their original sealing state after being punctured.

[0007] In a class of this embodiment, the main body comprises a cigarette holder, and the cigarette holder comprises an inhaling port; the first channel is disposed on the cigarette holder; the atomization core comprises an atomization tube and a heating core disposed in the atomization tube; the atomization tube comprises an atomization channel; the main body further comprises a first air passage; the first air passage communicates with the inhaling port via the atomization channel; the atomization channel is a part of the first air passage; the main body further comprises an air inlet corresponding to the second channel in position, and the air inlet communicates with the first air passage; the first air passage communicates with the second channel through the air inlet.

[0008] In a class of this embodiment, the main body further comprises a second air passage; the cigarette holder comprises an air outlet corresponding to the first channel in position; the air outlet communicates with the second air passage; the second air passage communicates with the first channel via the air outlet; and one end of the second air passage away from the air outlet

communicates with the first air passage.
[0009] In a class of this embodiment, the main body further comprises a first housing, and the storage chamber is disposed in the first housing; the atomization core is disposed in the storage chamber; the cigarette holder is disposed on one end of the first housing; one end of the first housing facing the cigarette holder is provided with a first sealing member for sealing the storage chamber; the first injection port is disposed on the first sealing member; the first injection port communicates with the first channel of the cigarette holder; and the first sealing film is disposed on the first sealing member.
[0010] In a class of this embodiment, the main body further comprises a second housing, and the first housing and the first sealing member are disposed in the first housing; the cigarette holder is disposed on one end of the second housing; a first clearance space is formed between the first sealing member and the first housing; the first sealing member comprises a first air hole; the first housing comprises an air tube; and the first air passage comprises the air tube, the first air hole, and the first clearance space which are mutually connected.
[0011] In a class of this embodiment, an airflow sensor is disposed on one end of the air tube away from the cigarette holder.
[0012] In a class of this embodiment, the second injection port is disposed on one end of the first housing away from the cigarette holder; a second sealing member is disposed around the second injection port; the second sealing film is disposed on the second sealing member; the main body further comprises a base; the base is disposed on the end of the first housing away from the cigarette holder; and the second channel is disposed on the base and corresponds to the second injection port.
[0013] In a class of this embodiment, the main body further comprises a bottom cover; the bottom cover is disposed on one end of the base away from the first housing; the base protrudes towards the bottom cover to form a guide tube; the second channel is disposed in the guide tube; the bottom cover comprises a second air hole corresponding to the second channel; the air inlet is a gap between an end part of the air tube and the bottom cover; a second clearance space is formed between the bottom cover and the base; a third clearance space is formed between the base and one end of the first housing; and the first air passage comprises the second clearance space, third clearance space, and the atomization channel which are mutually connected.
[0014] In a class of this embodiment, the first sealing film and the second sealing film have a thickness of 3.0-3.5 mm.
[0015] In a class of this embodiment, one end of the first channel away from the first sealing film is open, and one end of the inhaling port towards the cigarette holder is open. The open ends of the first channel and the inhaling port are towards one direction.
[0016] In a class of this embodiment, one end of the second channel away from the second sealing film is open; and detachable sealing components are disposed on one end of the first channel away from the first sealing film and on one end of the second channel away from the second sealing film, respectively.
[0017] In a class of this embodiment, the first housing further comprises an accommodation space, and a battery cell is disposed in the accommodation space.
[0018] In a class of this embodiment, the base further comprises a control panel; the control panel comprises two electrodes and conductive terminals electrically connected to the two electrodes, respectively; the heating core comprises an atomizable substance conductive element and an electric heating element; the electric heating element comprises a conductive wire electrically connected to the conductive terminals, and the conductive terminals is provided with an insulation sleeve for riveting and fixing the conductive wire with the conductive terminals; and the control panel comprises a power supply interface disposed on one side of the control panel away from the atomization core.
[0019] The following advantages are associated with the electronic cigarette of the disclosure. The

two ends of the main body of the electronic cigarette are provided with a first injection port and a second injection port, respectively, so that the atomizable substance can be injected into the electronic cigarette through the injection machine from top to bottom and from bottom to top, thus improving the production efficiency. The first and second channels serve as a guide for the injection needle of the injection machine, and the sealing film are not exposed, avoiding accidental punctures of the sealing film by the user and causing oil leakage. The first sealing film and the second sealing film are made of materials that are puncturable and able to restore to their original sealing state after being punctured, preventing the leakage of the atomizable substance, and no need to dismount the sealing films.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] FIG. **1** is a schematic diagram of an electronic cigarette in accordance with one embodiment of the disclosure;
[0021] FIG. **2** is a sectional view of an electronic cigarette in accordance with one embodiment of the disclosure;
[0022] FIG. **3** is an exploded view of an electronic cigarette in accordance with one embodiment of the disclosure;
[0023] FIG. **4** is a schematic diagram of an electronic cigarette in accordance with one embodiment of the disclosure in an angle of view;
[0024] FIG. **5** is a schematic diagram of an electronic cigarette in accordance with one embodiment of the disclosure in another angle of view;
[0025] FIG. **6** is an assembled diagram of a first housing and a first sealing member of an electronic cigarette in accordance with one embodiment of the disclosure;
[0026] FIG. **7** is an assembled diagram of a first housing and a second sealing member of an electronic cigarette in accordance with one embodiment of the disclosure;
[0027] FIG. **8** is an airflow diagram of a first air passage and a second air passage of an electronic cigarette in accordance with one embodiment of the disclosure; and
[0028] FIG. **9** is a sectional view of a first housing of an electronic cigarette in accordance with one embodiment of the disclosure.
DETAILED DESCRIPTION
[0029] To further illustrate the disclosure, embodiments detailing an electronic cigarette are described below. It should be noted that the following embodiments are intended to describe and not to limit the disclosure.
[0030] Atomizable substance of the disclosure means a mixture or auxiliary substance that can be atomized, in whole or in part, into an aerosol by an electronic or similar device. Atomizable substance can be an e-cigarette liquid, tobacco oil, a gel, a medical drug or skin lotion. By atomizing the medium, an aerosol can be produced and delivered to the user for vaping or absorption.
[0031] “Aerosol” means a colloidal dispersion system formed by the dispersion and suspension of small solid or liquid particles in a gaseous medium.
[0032] As shown in FIGS. **1**, **2**, **6** and **7**, an electronic cigarette of the disclosure comprises a main body **10** and an atomization core **16**.
[0033] The main body **10** comprises a storage chamber **52** for storing atomizable substance, and the atomization core **16** is disposed in the storage chamber **52**. The atomization core **16** is configured to heat the atomizable substance in the storage chamber **52** to produce aerosols. The main body **10** comprises a first channel **11***a* and a second channel **24***b* communicating with the storage chamber **52** for guiding and injection of the atomizable substance into the storage chamber. The first channel

11*a* comprises a first injection port **53** and the second channel **24***b* comprises a second injection port **54**. Both the first injection port and the second injection port communicate with the storage chamber. An injection needle can deliver the atomizable substance into the storage chamber **52** through the first injection port **53** and the second injection port **54** along the first channel **11***a* and the second channel **24***b*. The first and second channels serve as a guide for the injection needle of an injection machine, and the sealing film are not exposed, avoiding accidental punctures of the sealing film by the user and causing oil leakage.

[0034] In certain embodiment, the first channel **11***a* and the second channel **24***b* are respectively disposed on two opposite ends of the main body **10**, to adapt to different specifications of atomizable substance injection machines to meet the needs of different customers, avoiding the problem of slow production efficiency and high production costs caused by customers only having a single atomizable substance injection machine and needing to prepare different fixtures.

[0035] As shown in FIG. **2**, the first channel **11***a* is disposed on one end of the main body **10**. The second channel **24***b* and the first channel **11***a* are disposed on two opposite ends of the main body **10**, respectively. The arrangement mode of the first channel **11***a* and the second channel **24***b* is adapted to the common injection machines on the market, that is, injection machines that inject atomizable substance from top to bottom and injection machines that inject atomizable substance from bottom to top. In addition, the arrangement mode has a low processing cost and is easy to realize, can better adapt to existing electronic cigarettes, and does not occupy too much space of the storage chamber.

[0036] A first sealing film **131** is disposed in the first injection port **53** and a second sealing film **221** is disposed in the second injection port **54**, for sealing the first injection port **53** and the second injection port **54**, respectively. Furthermore, to facilitate the repeated injection of the atomizable substance, the first sealing film **131** and the second sealing film **221** are made of materials that are puncturable and able to restore to their original sealing state after being punctured, thus preventing the leakage of the atomizable substance. Optionally, the materials include but are not limited to silicone, rubber, or latex, such as fluororubber, nitrile rubber, or butadiene styrene rubber. In the embodiments of this disclosure, silicone is preferred, which has the advantages of being non-toxic and environmentally friendly.

[0037] In certain embodiments, the thickness of the first sealing film **131** and the second sealing film **221** ranges from 3.0 mm to 3.5 mm, including but not limited to 3.0 mm, 3.1 mm, 3.2 mm, 3.3 mm, 3.4 mm, and 3.5 mm. The thickness of the first sealing film **131** and the second sealing film **221** is the same or different, and those skilled in the art can adjust the parameter according to the volume size of the storage chamber **52** to ensure that it is easy to puncture and has good sealing performance after recovery without leakage. Preferably, the first sealing film **131** is 3.5 mm, and the second sealing film **221** is 3.0 mm.

[0038] In certain embodiments, the disclosure does not limit the quantity of the first injection port **53** and the second injection port **54**. Technical personnel in this field can decide the quantity as needed. In a preferred embodiment of this application, as shown in FIG. **2**, there are two first injection ports **53** and one second injection port **54**. By arranging more ports to adapt to different specifications of injection machines, the adaptability of the electronic cigarette is improved.

[0039] As shown in FIGS. **2** and **3**, the main body **10** comprises a first housing **14**, a second housing **11**, a base **23**, and a bottom cover **24**. One end of the second housing **11** is provided with a cigarette holder **111**, and the other end is opened. The first housing **14** and the base **23** are disposed inside the second housing **11**, and the storage chamber **52** is formed inside the first housing **14**. The base **23** is located at the end of the first housing **14** away from the cigarette holder to seal the storage chamber **52**. The bottom cover **24** is disposed at the opening of the second housing **11** for sealing the other end of the second housing **11**.

[0040] In certain embodiments, the main body **10** further comprises a third housing **12**. The third housing **12** is disposed around the second housing **11**, and the second housing **11** comprises a window corresponding to the storage chamber. A transparent cover plate is configured on the window to observe the amount of atomizable substance filled in the storage chamber. The third housing **12** is used to fix the cover plate on the periphery of the second housing **11**.

[0041] In certain embodiments, one end of the first housing **14** facing the cigarette holder is provided with a first sealing member **13** for sealing the storage chamber **52**. The first housing **14** and the first sealing member **13** are disposed inside the second housing **11**.

[0042] In certain embodiments, the first sealing member **13** is disposed between the first housing **14** and the second housing **11**. The inner wall of the first housing **14** is matched with the end of the first sealing member **13** facing the storage chamber **52** to form the storage chamber **52**. The storage chamber **52** is used to store the atomizable substance, and the atomization core **16** is accommodated in the storage chamber **52** of the first housing **14**.

[0043] In certain embodiments, the atomization core **16** is disposed in the storage chamber **52** to atomize the atomizable substance in the storage chamber **52**. Specifically, the atomization core **16** comprises an atomization tube **161**, a heating core **162**, and a support ring **163**; the heating core **162** and the support ring **163** are disposed in the atomization tube **161**. The atomization channel **55** is formed in the atomization tube **161**. The atomization tube **161** comprises a feeding hole, and the atomizable substance in the storage chamber **52** flows through the feeding hole to the heating core **162** for atomization. The support ring **163** is disposed on one side of the heating core **162** away from the storage chamber **52** for positioning the heating core **162** so that the heating core is facing to the feeding hole. As shown in FIG. **3**, the heating core **162** comprises an atomizable substance conductive element **162***a* and an electric heating element **162***b;* the atomizable substance conductive element **162***a* comprises an atomization surface and a liquid absorption surface opposite to the atomization surface, and the electric heating element **162***b* is arranged on the atomization surface of the atomizable substance conductive element **162***a.* The atomizable substance conductive element **162***a* is used to guide the atomizable substrate to the atomization surface, and the electric heating element **162***b* is used to heat the atomizable substrate to generate aerosols.

[0044] In certain embodiments, the atomizable substance conductive element **162***a* id made of flexible fibers such as cotton fibers, non-woven fabrics, glass fiber ropes, etc., or porous materials with microporous structures such as porous ceramics, so that the atomizable substance conductive element **162***a* can absorb and transfer the atomizable substance to the electric heating element **162***b* through internal voids or microporous structures. Correspondingly, the electric heating element **162***b* can be combined with the atomizable substance conductive element **162***a* through printing, deposition, sintering, or physical assembly, or wound outside the atomizable substance conductive element **162***a*, or wrapped inside the atomizable substance conductive element. In some embodiments, the suitable materials used for the electric heating element **162***b* include nickel, iron, stainless steel, nickel iron alloy, nickel chromium alloy, iron chromium aluminum alloy, or metallic titanium, or at least one of the conductive materials mentioned above, so that the atomizable substrate can be heated by resistance heating during conductivity, causing at least one component in the atomizable substrate to evaporate and form an aerosol.

[0045] In certain embodiments, the electric heating element **162***b* comprises multiple heating areas, which are interconnected to form a circular structure. In some examples, these heating areas can be connected in series or parallel in the circuit, or they can be partially connected in series and then in parallel in the circuit.

[0046] As shown in FIGS. **6**, **7**, **8** and **9**, a first clearance space **61** is formed between the first sealing member **13** and the first housing **14**; the first sealing member **13** comprises a first air hole **13***a;* the first housing **14** comprises an air tube **141**; and the first air passage Al comprises the air tube **141**, the first air hole **13***a*, and the first clearance space **61** which are mutually connected.

[0047] As shown in FIGS. **7**, **8** and **9**, the second injection port **54** is disposed on one end of the first housing **14** away from the cigarette holder **111**; a second sealing member **22** is disposed around the second injection port **54**; the second sealing film **221** is disposed on the second sealing member **22**; the base is disposed on the end of the first housing **14** away from the cigarette holder **111**; and the second channel **24***b* is disposed on the base **23** and corresponds to the second injection port **54**.

[0048] In certain embodiments, the first sealing member **13** and the second sealing member **22** are made of silicone, rubber, or latex, such as fluororubber, nitrile rubber, or styrene butadiene rubber.

[0049] In certain embodiments, the main body **10** further comprises an air inlet **23***a* corresponding to the second channel **24***b* in position, and the air inlet **23***a* communicates with the first air passage A**1**; the first air passage A**1** communicates with the second channel **24***b* through the air inlet **23***a* so as to obtain air from outside.

[0050] Further, as shown in FIGS. **2**, **3**, and **5**, the bottom cover **24** is disposed on one end of the base **23** away from the first housing **14**; the base **23** protrudes towards the bottom cover **24** to form a guide tube **231**; the second channel **24***b* is disposed in the guide tube **231**; the bottom cover **24** comprises a second air hole **24***a* corresponding to the second channel **24***b*. As shown in FIG. **8**, the air inlet **23***a* is a gap between an end part of the air tube **231** and the bottom cover **24**; a second clearance space **62** is formed between the bottom cover **24** and the base **23**; a third clearance space **63** is formed between the base **23** and one end of the first housing **14**; and the first air passage Al comprises the second clearance space **62**, third clearance space **63**, and the atomization channel **55** which are mutually connected.

[0051] As shown in FIG. **8**, an airflow sensor **27** is disposed on one end of the air tube **141** away from the cigarette holder **111**.

[0052] The atomization channel **55** is a part of the first air passage A**1**; the cigarette holder **111** comprises an air outlet **56** corresponding to the first channel **11***a* in position; the air outlet **56** communicates with the second air passage A**2**; the second air passage A**2** communicates with the first channel **11***a* via the air outlet **56**; and one end of the second air passage A**2** away from the air outlet **56** communicates with the first air passage A**1**. The air flows sequentially through the second channel **24***b*, the air inlet **23***a*, the first air passage A**1**, and the second air passage A**2**.

[0053] Specifically, as shown in FIGS. **6** and **7**, the cigarette holder **111** comprises an inhaling port **51** communicating with the atomization channel **55**. The main body **10** further comprises an air inlet **23***a* and an air outlet **56**. The main body **10** further comprises a first air passage A**1** and a second air passage A**2** communicating with the air inlet **23***a* and the air outlet **56**. An airflow sensor **27** is disposed at a joint of the first air passage A**1** and the second air passage A**2**. When smoking through the cigarette holder **111**, the air enters through the air inlet **23***a* and is divided into two parts, one of which flows out of the inhaling port **51** through the first air passage A**1**, and the other part flows through the second air passage A**2** and the air outlet **56**, and then flows out through the firs channel **11***a*, avoiding when a certain airflow channel is blocked, the airflow sensor **27** cannot be stimulated, leading to the electric heating element not being able to work properly.

[0054] In certain embodiments, the periphery of the airflow sensor **27** is equipped with a sealing silicone component **26** to enhance the airtightness of the airflow sensor **27**. The airflow sensor **27** is used to sense the changes in air pressure inside the main body **10** and output an electrical signal to excite the atomization core **16** for operation. Therefore, the airflow sensor **27** should have good airtightness to improve the sensitivity of the induction, and be used to fix the airflow sensor **27** to prevent displacement. Specifically, the sealing silicone component **26** wraps around the airflow sensor **27** and comprises a gas hole opposite to the airflow sensor **27**. The gas flows through the gas hole through the airflow sensor **27**, and the gas applies pressure to the airflow sensor **27**. The airflow sensor **27** senses the changes in air pressure and the control panel **21** drives the battery cell **15** and the atomization core **16** to work. The sealing silicone component **26** can be made of silicone, latex, or rubber.

[0055] As shown in FIG. **6**, in certain embodiments, the first housing further comprises an accommodation space **57**, and a battery cell **15** is disposed in the accommodation space **57**. The battery cell **15** is used to supply power to the electric heating element **162***b*. The voltage provided by the battery cell **15** is in the range of about 2.5 V to about 9.0 V, and the amperage of the current that the battery cell **15** can provide is in the range of about 2.5 A to about 20 A. The battery cell **15** can be any battery without restrictions.

[0056] In certain embodiments, as shown in FIG. **2**, the base **23** further comprises a control panel **21**; the control panel **21** comprises two electrodes and conductive terminals **20** electrically connected to the two electrodes, respectively; the heating core **162** comprises an atomizable substance conductive element **162***a* and an electric heating element **162***b;* the electric heating element **162***b* comprises a conductive wire electrically connected to the conductive terminals **20**, and the conductive terminals is provided with an insulation sleeve **19** for riveting and fixing the conductive wire with the conductive terminals; and the control panel **21** comprises a power supply interface **25** disposed on one side of the control panel away from the atomization core **16**. The power supply interface **25** charges the battery cell **15** through an external power source, and the battery cell **15** is electrically connected to the heating core **162** through the control panel for supplying power to the heating core **162**.

[0057] In certain embodiments, one end of the first channel away **11***a* from the first sealing film **131** is open, and one end of the inhaling port **51** towards the cigarette holder **111** is open. The two open ends are towards the same direction.

[0058] In certain embodiments, one end of the second channel **24***b* away from the second sealing film **221** is open; and detachable sealing components are disposed on one end of the first channel **11***a* away from the first sealing film **131** and on one end of the second channel **24***b* away from the second sealing film **221**, respectively. The sealing components are used to block the air inlet **23***a* and the air outlet **56**, to prevent the electronic cigarette from bumping and stimulating the airflow sensor **27** during transportation, causing the heating core **162** to work and burn out before the atomizable substance is injected into the storage chamber **52**. At the same time, the design can prevent the negative pressure change in the storage chamber during transportation or storage, causing the atomizable substance in the storage chamber to leak out from the electronic cigarette. Specifically, the sealing component is a sealing plug or a sealing sleeve. Understandably, the sealing plug is inserted into the injection channel for sealing, and the sealing sleeve is wrapped around one end of the electronic cigarette for sealing. The sealing plug can be made of silicone, latex, or rubber. The sealing sleeve can be made of silicone, latex, or rubber, or be a sticker, for example, to be attached to one end of the main body.

[0059] It will be obvious to those skilled in the art that changes and modifications may be made, and therefore, the aim in the appended claims is to cover all such changes and modifications.

Claims

1. An electronic cigarette, comprising: a main body, the main body comprising a storage chamber for storing atomizable substance and an atomization core for heating the atomizable substance; wherein: the main body further comprises a first end comprising a first channel for guiding and injection of the atomizable substance, and a second end comprising a second channel for guiding and injection of the atomizable substance; the first end is opposite to the second end; the first channel comprises a first injection port for the atomizable substance and the second channel comprises a second injection port for the atomizable substance; both the first injection port and the second injection port communicate with the storage chamber; a first sealing film is disposed in the first injection port and a second sealing film is disposed in the second injection port, and the first sealing film and the second sealing film are puncturable and able to restore to their original sealing state after being punctured.

2. The electronic cigarette of claim 1, wherein the main body comprises a cigarette holder, and the cigarette holder comprises an inhaling port; the first channel is disposed on the cigarette holder; the atomization core comprises an atomization tube and a heating core disposed in the atomization tube; the atomization tube comprises an atomization channel; the main body further comprises a first air passage; the first air passage communicates with the inhaling port via the atomization channel; the atomization channel is a part of the first air passage; the main body further comprises an air inlet corresponding to the second channel in position, and the air inlet communicates with the first air passage; the first air passage communicates with the second channel through the air inlet.

3. The electronic cigarette of claim 2, wherein the main body further comprises a second air passage; the cigarette holder comprises an air outlet corresponding to the first channel in position; the air outlet communicates with the second air passage; the second air passage communicates with the first channel via the air outlet; and one end of the second air passage away from the air outlet communicates with the first air passage.

4. The electronic cigarette of claim 3, wherein the main body further comprises a first housing, and the storage chamber is disposed in the first housing; the atomization core is disposed in the storage chamber; the cigarette holder is disposed on one end of the first housing; one end of the first housing facing the cigarette holder is provided with a first sealing member for sealing the storage chamber; the first injection port is disposed on the first sealing member; the first injection port communicates with the first channel of the cigarette holder; and the first sealing film is disposed on the first sealing member.

5. The electronic cigarette of claim 4, wherein the main body further comprises a second housing, and the first housing and the first sealing member are disposed in the first housing; the cigarette holder is disposed on one end of the second housing; a first clearance space is formed between the first sealing member and the first housing; the first sealing member comprises a first air hole; the first housing comprises an air tube; and the first air passage comprises the air tube, the first air hole, and the first clearance space which are mutually connected.

6. The electronic cigarette of claim 5, wherein an airflow sensor is disposed on one end of the air tube away from the cigarette holder.

7. The electronic cigarette of claim 4, wherein the second injection port is disposed on one end of the first housing away from the cigarette holder; a second sealing member is disposed around the second injection port; the second sealing film is disposed on the second sealing member; the main body further comprises a base; the base is disposed on the end of the first housing away from the cigarette holder; and the second channel is disposed on the base and corresponds to the second injection port.

8. The electronic cigarette of claim 7, wherein the main body further comprises a bottom cover; the bottom cover is disposed on one end of the base away from the first housing; the base protrudes towards the bottom cover to form a guide tube; the second channel is disposed in the guide tube; the bottom cover comprises a second air hole corresponding to the second channel; the air inlet is a gap between an end part of the air tube and the bottom cover; a second clearance space is formed between the bottom cover and the base; a third clearance space is formed between the base and one end of the first housing; and the first air passage comprises the second clearance space, the third clearance space, and the atomization channel which are mutually connected.

9. The electronic cigarette of claim 1, wherein the first sealing film and the second sealing film have a thickness of 3.0-3.5 mm.

10. The electronic cigarette of claim 2, wherein one end of the first channel away from the first sealing film is open, and one end of the inhaling port towards the cigarette holder is open; and open ends of the first channel and the inhaling port are towards one direction.

11. The electronic cigarette of claim 2, wherein one end of the second channel away from the second sealing film is open; and detachable sealing components are disposed on one end of the first channel away from the first sealing film and on the one end of the second channel away from the second sealing film, respectively.

12. The electronic cigarette of claim 1, wherein the first housing further comprises an accommodation space, and a battery cell is disposed in the accommodation space.

13. The electronic cigarette of claim 7, wherein the base further comprises a control panel; the control panel comprises two electrodes and conductive terminals electrically connected to the two electrodes, respectively; the heating core comprises an atomizable substance conductive element and an electric heating element; the electric heating element comprises a conductive wire electrically connected to the conductive terminals, and the conductive terminals is provided with an insulation sleeve for riveting and fixing the conductive wire with the conductive terminals; and the control panel comprises a power supply interface disposed on one side of the control panel away from the atomization core.
